

Practice — Thinking Like a Scientist

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

Which of these is a **scientific (testable) question**?

- (a) Are puris tasty? (b) Does thicker dough take longer to puff up?
(c) Which is the best food in the world? (d) Should everyone eat puris?

Q2 • MCQ**[1]**

"The leaf is green." This statement is an example of an —

- (a) observation (b) inference (c) prediction (d) conclusion

Q3 • MCQ**[1]**

Which is a **quantitative** observation?

- (a) The oil smelled strong. (b) The puri turned golden. (c) The puri puffed up in 8 seconds.
(d) The dough felt soft.

Q4 • ASSERTION–REASON**[1]**

A: "The puri puffed up because steam formed inside" is an inference.

R: An inference is an explanation worked out from observations.

- (a) Both A and R true, and R explains A (b) Both true, but R does not explain A
(c) A true, R false (d) A false, R true

Q5 • TRUE / FALSE**[1]**

You need a proper laboratory to carry out a scientific investigation.

- (a) True (b) False

Q6 · FILL IN THE BLANK

[1]

An observation described with a number and a unit is called a _____ observation.

Q7 · FILL IN THE BLANK

[1]

The two textbook symbols are the root (grounded in observation) and the _____ (curiosity taking flight).

Short answer (2 marks each)

Q8

[2]

Define *observation* and *inference*, giving one example of each.

Q9

[2]

Give one qualitative and one quantitative observation you could make while a phulka swells on a flame.

Q10

[2]

Rewrite the vague question "Why are some puris better?" as a scientific question that can be tested.

Long / application (3 marks each)

Q11

[3]

Your bicycle tyre goes flat overnight. Describe how you would investigate this **like a scientist** — state your question, one thing you would change, and one thing you would observe or measure.

Q12

[3]

Pick out **two observations** and **one inference**: "I dropped the dough into the oil. It sank, then rose and puffed up in 6 seconds. The bottom side was thinner, probably because it touched the hot pan first."

[Check yourself](#) → Answer key